Bryce Wedig

Email: bryce.wedig@gmail.com
ORCiD: 0000-0002-0748-7312
GitHub: github.com/bryce-wedig
Website: bryce-wedig.github.io

Research Interests

Visual and near-infrared space telescopes (extragalactic astronomy with, survey and image simulation, imaging data reduction and analysis), strong gravitational lensing, dark matter

Education

Washington University in St. Louis — St. Louis, MO, USA
Ph.D. in Physics — 2022–Present (Expected September 2027)
- A.M. in Physics (en route) achieved May 2024, GPA: 3.95/4.00

Trinity College, University of Cambridge — Cambridge, UK
M.Phil. in History and Philosophy of Science and Medicine, *First Class Honours* — 2018–2019

Kenyon College — Gambier, OH, USA
B.A. in Physics, Philosophy, *magna cum laude* — 2014–2018
- GPA: 3.89/4.00 (Physics 3.96/4.00, Philosophy 4.00/4.00)

Lady Margaret Hall, University of Oxford — Oxford, UK
Visiting Student Programme, Physics and Philosophy — 2016–2017
- GPA: 4.00/4.00

Honors and Awards

- **1st Place, Student E-Poster Competition (Physical Sciences category)** — 2026
  2026 Annual Meeting of the American Association for the Advancement of Science
- **1st Place, Poster Competition**, Arizona Winter School on Multi-Probe Cosmology in the Roman ST Era — 2026
- **1st Place (Graduate Division)**, Washington University Physics Research Symposium — 2025
- **1st Place (Graduate Division)**, Washington University Physics Research Symposium — 2024
- **Future Investigators in NASA Earth and Space Science and Technology (FINESST)** — 2024
  Maximizing JWST and Roman Dark Matter Science with Strong Gravitational Lensing
  $150,000 award over three years
- **First Class (Distinction)**, M.Phil. Dissertation, University of Cambridge — 2019
- **Member, Phi Beta Kappa**, Beta of Ohio, Kenyon College — 2018
- **Distinction**, Senior Exercise in Physics, Department of Physics, Kenyon College — 2018
- **Distinction**, Senior Exercise in Philosophy, Department of Philosophy, Kenyon College — 2018
- **Member, Sigma Pi Sigma**, the Physics Honor Society (American Institute of Physics) — 2017
- **Merit List Scholar**, Kenyon College — 2014–2018
- **Trustee Opportunity Scholarship**, Kenyon College — 2014–2018

Accepted Proposals

- **Co-Investigator**, *Roman* Cycle 1 (Program ID: 19045). *All the strong lenses in Roman HLWAS*. PI: Simon Birrer

Publications

1. Khadka, N. *et al.* SLSim: a strong lensing population simulation package. *arXiv e-prints,* arXiv:2603.17138. arXiv: 2603.17138 [astro-ph.CO] (Mar. 2026).
2. Sharma, P. *et al.* Strong Lensing Tomography: Double and pseudo multi-source plane strong gravitational lensing to constrain dark energy. *arXiv e-prints,* arXiv:2607.01005. arXiv: 2607.01005 [astro-ph.CO] (July 2026).
3. **Wedig, B.** *et al.* The Roman View of Strong Gravitational Lenses. *The Astrophysical Journal* **986,** 42. arXiv: 2506.03390 [astro-ph.CO] (June 2025).

Professional Experience

- **Technical Consultant**, R&D Technical Services, Veeva Systems — Oct 2021–Jun 2022
  Developed and implemented custom software integrations with Veeva Vault and extended system functionality utilizing Vault Java SDK and REST API, Amazon Web Services. Managed integration and development projects.
- **Associate Consultant**, R&D Services, Veeva Systems — Oct 2019–Oct 2021
  Implemented Veeva's Vault cloud software applications at global life sciences companies. Led candidate case study interviews and mentored new hires' mock projects.

Selected Presentations

Invited

- "The Roman View of Strong Gravitational Lenses" — Dec 2025
  "Nature of Dark Matter at Small Scales" Virtual Seminar Series

Contributed

- "A JWST Survey of Subhalos Across Cosmic Time" — Oct 2025
  1st Workshop Scaling up Lensing, University in Liège, Liège, Belgium
- "The Microphysics of Dark Matter with the Habitable Worlds Observatory" — Jul 2025
  Towards the Habitable Worlds Observatory: Visionary Science and Transformational Technology Washington, D.C., USA
- "The Roman View of Strong Gravitational Lenses" — Jul 2024
  Challenging Theory with Roman: From Planet Formation to Cosmology, Pasadena, CA, USA

Mentoring

- **Jenna Eggerding**, Undergraduate student at Washington University in St. Louis — March 2026–Present
  Modeling strong lenses from JWST. Plan to present results at Fall 2026 WashU Undergraduate Research Symposium.
- **Jodie Xiao**, Undergraduate student at Washington University in St. Louis — May 2025–Present
  Participated in the LSST Strong Lens Data Challenge in Fall 2025. Trained a convolutional neural network to classify strong lenses, achieved score 98.1/100 placing 9th out of 21 teams. Presented results at Fall 2025 WashU Undergraduate Research Symposium and 2025 Washington University Physics Research Symposium.

Service and Public Engagement

- **Speaker**, "Science in St. Louis" Public Seminar Series — May 2026
  "Dark Matter Detectives: Using NASA's Space Telescopes to See the Invisible Universe"
  Academy of Science of St. Louis
- **Contributor**, "ExoCore: An Open Science Curriculum for Exoplanet Research" — Dec 2024–Present
  NASA Open Science Initiative
- **Co-president**, Washington University ProSPER — 2024–2026
  Science communication graduate student group

Presentations

Invited Talks

- "The Roman View of Strong Gravitational Lenses" Dec 2025
  "Nature of Dark Matter at Small Scales" Virtual Seminar Series

Contributed Talks

- "Probing Dark Matter Microphysics with JWST's Galaxy-Galaxy Strong Gravitational Lenses" Dec 2025
  53rd Mid-American Regional Astrophysics Conference, Columbia, MO, USA
- "A JWST Survey of Subhalos Across Cosmic Time" Oct 2025
  1st Workshop Scaling up Lensing, University in Liège, Liège, Belgium
- "The Microphysics of Dark Matter with the Habitable Worlds Observatory" Jul 2025
  Towards the Habitable Worlds Observatory: Visionary Science and Transformational Technology Washington, D.C., USA
- "Hunting for Dark Matter with NASA's Next Space Telescope" Apr 2025
  2025 Graduate Research Symposium, Washington University in St. Louis
- "The Roman View of Strong Gravitational Lenses" Jan 2025
  245th Meeting of the American Astronomical Society, National Harbor, MD, USA
- "The Roman View of Strong Gravitational Lenses" Dec 2024
  52nd Mid-American Regional Astrophysics Conference, Lawrence, KS, USA
- "The Roman View of Strong Gravitational Lenses" Jul 2024
  Challenging Theory with Roman: From Planet Formation to Cosmology, Pasadena, CA, USA
- "Image Simulation of Strong Gravitational Lenses Detectable by the Roman Space Telescope" Jan 2024
  243rd Meeting of the American Astronomical Society, New Orleans, LA, USA

Seminars

- "Searching for dark matter substructure in JWST's galaxy-scale strong lenses" Nov 2025
  Washington University Cosmology and Astroparticle Physics Meeting
  Department of Physics, Washington University in St. Louis
- "Hunting for Dark Matter with Space Telescopes" Feb 2025
  A&S Research Roundtable, School of Arts & Sciences, Washington University in St. Louis
- "Maximizing JWST and Roman Dark Matter Science with Strong Gravitational Lensing" Nov 2024
  Washington University Cosmology and Astroparticle Physics Meeting
  Department of Physics, Washington University in St. Louis
- "The Roman View of Strong Gravitational Lenses" Nov 2024
  Physics Graduate Student Seminar, Department of Physics, Washington University in St. Louis
- "The Roman View of Strong Gravitational Lenses" Apr 2024
  Washington University Cosmology and Astroparticle Physics Meeting
  Department of Physics, Washington University in St. Louis
- "Investigating Dark Matter Substructure with Gravitational Lensing" Nov 2023
  Physics Graduate Student Seminar, Department of Physics, Washington University in St. Louis
- "Lessons for wavefunction ontology from Koopman-von Neumann theory" Easter 2019
  MPhil Seminar, Department of History and Philosophy of Science, University of Cambridge
- "Assessing the Quantum Liar Paradox" Lent 2019
  MPhil Seminar, Department of History and Philosophy of Science, University of Cambridge
- "Is Nature Nonlocal?" Apr 2018

Senior Exercise, Department of Physics, Kenyon College

- "Reducing quantum noise in LIGO: Characterization of an ultra-low loss polarizing beam splitter" Aug 2016
  Ole Miss Chemistry Summer Research Program, University of Mississippi

Poster Presentations

- "Forecasting Dark Matter Subhalo Detections with NASA's Habitable Worlds Observatory" Feb 2026
  2026 Annual Meeting of the American Association for the Advancement of Science
  Phoenix, AZ
- "Forecasting Dark Matter Subhalo Detections with NASA's Habitable Worlds Observatory" Nov 2025
  Washington University Physics Research Symposium
  Department of Physics, Washington University in St. Louis
- "Hunting for Dark Matter with NASA's Next Space Telescope" Apr 2025
  50th Anniversary Symposium, McDonnell Center for the Space Sciences
  Washington University in St. Louis
- "Hunting for Dark Matter with NASA's Next Space Telescope" Apr 2025
  2025 Imaging Science Pathway Retreat, Washington University in St. Louis
- "The Roman View of Strong Gravitational Lenses" Nov 2024
  Washington University Physics Research Symposium
  Department of Physics, Washington University in St. Louis
- "Image Simulation of Strong Gravitational Lenses Detectable by the Roman Space Telescope" Nov 2023
  Washington University Physics Research Symposium
  Department of Physics, Washington University in St. Louis

Teaching

- **Guest Lecturer**, Astrostatistics (PHYS 4680/5680) Spring 2025
  Department of Physics, Washington University in St. Louis
- **Guest Lecturer**, Ampersand: Gateway Expeditions into Exoplanets (PHYS 1210) Fall 2024
  Department of Physics, Washington University in St. Louis
- **Assistant in Instruction**, Physics I Laboratory (PHYS 191L) Fall 2023
  Department of Physics, Washington University in St. Louis
- **Assistant in Instruction**, Physics II Laboratory (PHYS 192L) Spring 2023
  Department of Physics, Washington University in St. Louis
- **Apprentice Teacher**, Intensive Introductory Japanese (JAPN 111-112Y) Fall 2017
  Department of Modern Languages and Literatures, Kenyon College
- **Teaching Assistant**, Advanced Topics in Physics: Special Relativity (SREL) Summer 2017
  Johns Hopkins Center for Talented Youth
- **Tutor**, Oscillations and Waves (PHYS 245) Spring 2016
  Department of Physics, Kenyon College
- **Apprentice Teacher**, Intermediate Japanese (JAPN 213-214Y) Fall 2015–Spring 2016
  Department of Modern Languages and Literatures, Kenyon College
- **Apprentice Teacher**, Intensive Introductory Japanese (JAPN 111-112Y) Fall 2014–Spring 2015
  Department of Modern Languages and Literatures, Kenyon College

Workshops and Schools

- Arizona Winter School on Multi-Probe Cosmology in the Roman ST Era — Jan 2026
  University of Arizona
- JWST Summer School on High-Redshift Transients — Aug 2025
  Space Telescope Science Institute
- ALMA Cycle 12 Proposal Preparation Workshop — Apr 2025
  University of Missouri